A First Course in Abstract Algebra

Section 50: Splitting Fields

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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50 Splitting Fields

Exercise 50.1: Splitting a quadratic

The roots of x^2+3 are $\pm\sqrt{-3}$. Thus the splitting field is $\mathbb{Q}(\sqrt{-3})$ of degree 2, since the polynomial has no rational root.

Exercise 50.2: Fourth roots of unity

The roots of x^4-1 are $1,-1,i,-i$. Adjoining i suffices and is necessary. The splitting field $\mathbb{Q}(i)$ has degree 2 over $\mathbb{Q}$.

Exercise 50.3: Two distinct quadratic radicals

The splitting field is $\mathbb{Q}(\sqrt{2},\sqrt{3})$. It has degree 4: if $\sqrt{3}=a+b\sqrt{2}$ with rational a,b, squaring would give $3=a^2+2b^2+2ab\sqrt{2}$, hence $ab=0$. The alternatives would require either a rational square root of 3 or a rational square root of $3/2$, both impossible. Thus the second quadratic doubles the first degree.

Exercise 50.4: Splitting an irreducible cubic

Let $r=\sqrt[3]{3}$ and ω be a primitive cube root of unity. The splitting field is $\mathbb{Q}(r,\omega)$. Eisenstein at 3 gives $[\mathbb{Q}(r):\mathbb{Q}]=3$, and ω is nonreal while $\mathbb{Q}(r)\subset\mathbb{R}$. Its quadratic polynomial therefore doubles the degree, giving 6.

Exercise 50.5: Third roots of unity

Factor $x^3-1=(x-1)(x^2+x+1)$. The quadratic has discriminant -3, so its splitting field is $\mathbb{Q}(\sqrt{-3})=\mathbb{Q}(\omega)$, of degree 2.

Exercise 50.6: Combining square and cube roots

The roots require $\sqrt{2}, r=\sqrt[3]{2}$, and ω. Put $s=2^{1/6}$; then $s=\sqrt{2}/r$, while $s^2=r$ and $s^3=\sqrt{2}$. Thus the splitting field is $\mathbb{Q}(s,\omega)$. Eisenstein makes x^6-2 irreducible, so $[\mathbb{Q}(s):\mathbb{Q}]=6$. This real field does not contain the nonreal ω, giving total degree 12.

Exercise 50.7: Automorphisms of the real cubic field

The order is 1. An automorphism sends the real $r=\sqrt[3]{2}$ to a root of x^3-2 inside the real field $\mathbb{Q}(r)$. Its only real root is r, so every automorphism fixes this generator and is the identity.

Exercise 50.8: Automorphisms of its splitting field

The splitting field $E = \mathbb{Q}(r, i\sqrt{3})$ has degree 6 over $\mathbb{Q}$. Characteristic zero gives six embeddings, all of which are automorphisms because E is a splitting field. Thus the group order is 6.

Exercise 50.9: Fixing the real cubic generator

The extension $E/\mathbb{Q}(r)$ has degree 2, obtained by adjoining $i\sqrt{3}$. Its two automorphisms fix r and send $i\sqrt{3}$ to itself or its negative. Hence the relative group has order 2.

Exercise 50.10: A cubic splits in an eight-element field

The polynomial $f = x^3 + x^2 + 1$ has no root in $\mathbb{Z}_2$, so is irreducible. Thus $\mathbb{Z}_2(\alpha)$ has eight elements. Frobenius preserves roots of polynomials over $\mathbb{Z}_2$, so $\alpha, \alpha^2, \alpha^4$ are roots. They are distinct: a nonzero nonidentity element of the group of seven nonzero elements has order 7, and the exponents 1,2,4 are distinct modulo 7. Hence these exhaust its three roots and f splits there. From $\alpha^3 = \alpha^2 + 1$, the last root is $\alpha^4 = \alpha^2 + \alpha + 1$.

Exercise 50.11: Correcting the splitting-field criterion

In the stated condition, "every polynomial" must be replaced by "every irreducible polynomial." Otherwise even the splitting field $E = F = \mathbb{Q}$ fails the proposed condition for $(x - 1)(x^2 - 2)$. An algebraic extension E/F is a splitting field of a family of polynomials precisely when each irreducible polynomial over F with one zero in E has all its zeros in E.

Exercise 50.12: Correcting the meaning of splitting

A polynomial splits over E when it is a nonzero constant times a product of *linear* polynomials in $E[x]$. Merely factoring into lower-degree factors is insufficient: $(x^2 - 2)(x^2 - 3)$ factors over $\mathbb{Q}$ but does not split there.

Exercise 50.13: Bounds for a splitting-field degree

For a nonconstant degree-n polynomial, $1 \leq [E : F] \leq n!$. Adjoin a first root of degree at most n, factor out its linear factor, then adjoin a root of the remaining factor of degree at most $n - 1$, and continue. The tower law bounds the product by $n(n - 1) \cdots 1$. No irreducibility assumption gives a stronger positive lower bound: a polynomial already split over F has degree-one splitting field. For a nonzero constant polynomial the splitting field is F as well.

Exercise 50.14: Ten statements about splitting fields

(a) True. A conjugation isomorphism extends to the closure, and normality preserves E; the converse follows by applying an automorphism to a minimal polynomial.

(b) False. $\mathbb{R}/\mathbb{Q}$ is not algebraic, whereas a splitting field is algebraic.

(c) True: the trivial extension is the splitting field of x.

(d) True: $\mathbb{C}$ splits x^2+1 over $\mathbb{R}$ and is generated by its roots.

(e) True: it splits x^2+1 over $\mathbb{Q}$.

(f) True: $\mathbb{Q}(\pi)$ splits $x^2-\pi^2$ over $\mathbb{Q}(\pi^2)$.

(g) False. An isomorphism need not fix F or even have its image in the same field; send the transcendental generator of $E=F=\mathbb{Q}(t)$ to an independent indeterminate.

(h) False. For $E=F=\mathbb{Q}(t)$, the embedding $t\mapsto t^2$ has image $\mathbb{Q}(t^2)\subsetneq E$ and lies inside an algebraic closure of F.

(i) True. This is the defining embedding property of a normal algebraic extension.

(j) True. An algebraic closure is generated by all roots of all polynomials over F.

Exercise 50.15: Why irreducibility is essential

Take $F=E=\mathbb{Q}$, a splitting field of x over $\mathbb{Q}$. The reducible polynomial $(x-1)(x^2-2)$ has the zero 1 in E but its other two zeros are $\pm\sqrt{2}$, outside E. Thus the corollary about an irreducible polynomial having one root in a splitting field does not extend to reducible polynomials.

Exercise 50.16: Group orders in a tower

The intended multiplicativity formula has $|G(K/F)|$ on the left; the printed formula repeats $G(E/F)$ there.

(a) No. For $F=\mathbb{Q}$, $E=\mathbb{Q}(\sqrt[3]{2})$ and $K=\mathbb{Q}(\sqrt[3]{2},i\sqrt{3})$, the group orders are 6,1,2 respectively. Thus $6\neq 2\cdot 1$.

(b) Yes when the relevant fields are splitting fields over the bottom field. Their group orders equal their embedding indices, including for K/E, since a normal extension over F is normal over every intermediate field. The multiplicativity of indices then gives $|G(K/F)|=|G(K/E)||G(E/F)|$. This argument does not require separability.

Exercise 50.17: One polynomial suffices for a finite normal extension

Choose finitely many generators $\alpha_1,\ldots,\alpha_m$ for E/F and let p_j be their minimal polynomials over F. Since E is normal, each p_j splits in E. Let $f=\prod_j p_j$. Its splitting field inside E contains every α_j, so contains E; it is also contained in E, since all its roots are there. Hence it equals E.

Exercise 50.18: A quadratic extension is a splitting field

Choose $\alpha\in E\setminus F$. The tower law gives $E=F(\alpha)$ and a degree-two minimal polynomial x^2+bx+c. Its other root, counted with multiplicity, is $-b-\alpha\in E$. Thus the polynomial splits in E, and one root generates E, proving the claim even in characteristic 2 (where the roots may coincide).

Exercise 50.19: Containing all conjugates

If E/F is a splitting field, every irreducible polynomial with a root in E splits there, so all conjugates of every element belong to E. Conversely, for each $\alpha \in E$, take its minimal polynomial. The hypothesis places all its roots in E. The splitting field of this family is contained in E and contains every α, hence is exactly E.

Exercise 50.20: No nontrivial automorphism of the real cubic

Every automorphism of $\mathbb{Q}(r)$ fixes $\mathbb{Q}$, and the image of $r = \sqrt[3]{2}$ must satisfy $x^3 - 2 = 0$. The other two roots $r\omega, r\omega^2$ are nonreal, whereas the field is real. Thus r is fixed. Since it generates the field, the automorphism is the identity.

Exercise 50.21: The cyclic subgroup fixing the quadratic radical

Put $r = \sqrt[3]{2}$ and $\omega = (-1 + i\sqrt{3})/2$. The field $\mathbb{Q}(i\sqrt{3}) = \mathbb{Q}(\omega)$ has degree 2 over $\mathbb{Q}$, so its relative degree in the degree-six splitting field is 3. The map $\rho(r) = r\omega$, $\rho(\omega) = \omega$ is an automorphism, and its powers send r successively to $r\omega^2$ and r. It has order 3 and accounts for all three relative embeddings. Thus the group is cyclic, isomorphic to $(\mathbb{Z}_3, +)$.

Exercise 50.22: The permutation representation on roots

(a) For an automorphism *fixing* F, $f(\sigma(\alpha)) = \sigma(f(\alpha)) = 0$. Its inverse shows that this gives a permutation of the roots. The base-field qualification is necessary.

(b) The roots generate the splitting field over F. Agreement on all roots and on F implies agreement on every rational expression in them, hence on the whole field.

(c) Restriction to the roots is therefore an injective homomorphism $G(E/F) \hookrightarrow S_\Omega$, where Ω is the set of distinct roots. Its image is the subgroup of permutations that preserve all algebraic relations among the roots over F.

Exercise 50.23: The full cubic permutation group

(a) The field $E = \mathbb{Q}(r, \omega)$ has degree 6 and is separable in characteristic zero. All six rational embeddings are automorphisms, so $|G(E/\mathbb{Q})| = 6$.

(b) Its action on the three roots gives an injection into S_3. As both groups have six elements, the injection is onto, proving $G(E/\mathbb{Q}) \cong S_3$.

Exercise 50.24: A prime-root-of-unity extension

For a prime p, every nonidentity pth root ζ is primitive, and its powers give all roots of $x^p - 1$. Thus its splitting field is $\mathbb{Q}(\zeta)$. The minimal polynomial $1 + x + \cdots + x^{p-1}$ is irreducible over $\mathbb{Q}$ by the prime cyclotomic theorem, so the extension degree is $p - 1$ (including degree 1 for $p = 2$).

Exercise 50.25: Uniqueness of splitting fields in different closures

There is an isomorphism $\tau : \overline{F} \to \overline{F}'$ fixing F. It maps roots of f to roots of f because the coefficients are fixed. Its inverse gives the reverse inclusion, so it bijects the two full root sets. Since each splitting field is generated by the respective roots, $\tau(E) = E'$. The restriction is the desired F-isomorphism.